Check for updates

Circulation

ORIGINAL RESEARCH ARTICLE

Olmesartan Restores *LMNA* Function in Haploinsufficient Cardiomyocytes

Eric J. Kort, MD, MS*; Nazish Sayed, MD, PhD*; Chun Liu, PhD; Gema Mondéjar-Parreño, PhD; Jens Forsberg, PhD; Emily Eugster, MS; Sean M. Wu, MD, PhD; Joseph C. Wu, MD, PhD; Stefan Jovinge, MD, PhD

BACKGROUND: Gene mutations are responsible for a sizeable proportion of cases of heart failure. However, the number of patients with any specific mutation is small. Repositioning of existing US Food and Drug Administration–approved compounds to target specific mutations is a promising approach to efficient identification of new therapies for these patients.

METHODS: The National Institutes of Health Library of Integrated Network-Based Cellular Signatures database was interrogated to identify US Food and Drug Administration–approved compounds that demonstrated the ability to reverse the transcriptional effects of *LMNA* knockdown. Top hits from this screening were validated in vitro with patient-specific induced pluripotent stem cell–derived cardiomyocytes combined with force measurement, gene expression profiling, electrophysiology, and protein expression analysis.

RESULTS: Several angiotensin receptor blockers were identified from our in silico screen. Of these, olmesartan significantly elevated the expression of sarcomeric genes and rate and force of contraction and ameliorated arrhythmogenic potential. In addition, olmesartan exhibited the ability to reduce phosphorylation of extracellular signal–regulated kinase 1 in *LMNA*-mutant induced pluripotent stem cell–derived cardiomyocytes.

CONCLUSIONS: In silico screening followed by in vitro validation with induced pluripotent stem cell–derived models can be an efficient approach to identifying repositionable therapies for monogenic cardiomyopathies.

Key Words: cardiomyopathies ■ drug repositioning ■ gene expression profiling ■ lamins

Editorial, see p 1449

The cost of drug development from initial concept to US Food and Drug Administration (FDA) approval has been estimated to be about US $2.6 billion.[1] This cost precludes the development of targeted therapies for rare diseases such as monogenetic cardiomyopathies. As part of the Library of Integrated Network-Based Cellular Signatures (LINCS) program funded by the National Institutes of Health, the Broad Institute of the Massachusetts Institute of Technology has publicly released transcriptional profiles quantifying the effects of >25 000 perturbagens on the expression of 978 genes in up to 77 cell lines.[2] Transcriptomics has been shown to be a powerful tool in repurposing drugs,[3,4] and this data set provides us with the unique opportunity to systematically identify small-molecule mimics or inhibitors of specific genes, thereby identifying novel treatments for genetic disorders. In this report, we take this approach to identify a novel drug therapy for a monogenic form of familial dilated cardiomyopathy combined with the transcriptional profile of FDA-approved drugs. This approach could potentially be replicated for a wide range of monogenic diseases.

Correspondence to: Stefan Jovinge, MD, PhD, Skåne University Hospital, Department of Research, Education, Development and Innovation, Lasarettsgatan 40A, 221 85 Lund, Sweden. Email stefan.jovinge@skane.se
*E.J. Kort and N. Sayed contributed equally.
Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.121.058621.
For Sources of Funding and Disclosures, see page 1447.
© 2025 The Authors. *Circulation* is published on behalf of the American Heart Association, Inc., by Wolters Kluwer Health, Inc. This is an open access article under the terms of the Creative Commons Attribution Non-Commercial-NoDerivs License, which permits use, distribution, and reproduction in any medium, provided that the original work is properly cited, the use is noncommercial, and no modifications or adaptations are made.
Circulation is available at www.ahajournals.org/journal/circ

Downloaded from http://ahajournals.org by on February 4, 2026

Clinical Perspective

What Is New?

- This study demonstrates that large catalogs of gene expression data can be used to match the effects of gene mutations to the effects of drugs to identify novel, targeted treatment options.
- The study suggests that olmesartan specifically targets the pathophysiology of heart failure attributable to *LMNA* mutation and may be uniquely beneficial to this patient population relative to other common heart failure drugs.

What Are the Clinical Implications?

- Olmesartan may provide targeted, mutation-specific therapy for patients with heart failure with *LMNA* mutations that is superior to other options.
- Patients with diseases caused by other gene mutations may benefit from drug discovery using a similar approach.

Nonstandard Abbreviations and Acronyms

CM	cardiomyocyte
CMAP	connectivity map
DMSO	dimethyl sulfoxide
EHT	engineered heart tissue
ERK	extracellular signal–regulated kinase
FDA	US Food and Drug Administration
FP	field potential
FPD	field potential duration
iPSC	induced pluripotent stem cell
LINCS	Library of Integrated Network-Based Cellular Signatures
MEA	multielectrode array
PDGF	platelet-derived growth factor

Mutations in the *LMNA* gene are associated with familial dilated cardiomyopathy[5–7] and premature aging syndromes.[8] *LMNA* encodes for lamin A/C, a nuclear structural protein that also plays a key role in chromatin organization and transcriptional regulation.[9] These mutations may be either missense mutations, leading to severe early disease presumably attributable to dominant negative or pathological gain of function, or nonsense mutations characterized by late-onset disease attributable to haploinsufficiency.[10,11] *LMNA* has been shown to be involved in the regulation of several signaling pathways whose disruption may contribute to the disease phenotype. These include platelet-derived growth factor (PDGF) signaling, autophagy through Akt/Beclin signaling, mitogen-activated protein kinase signaling, and histone modification.[12–16]

In this study, we leveraged analysis of the LINCS L1000 data set combined with induced pluripotent stem cell (iPSC)–based in vitro assays to identify FDA-approved compounds that can reverse the effects of *LMNA* mutation at the transcriptional and functional levels (Figure 1).

METHODS

Cells used for many of the analyses performed were obtained from human donors, and the protocol for isolation and the use of these patient blood-derived peripheral blood mononuclear cell were approved by the Stanford University Human Subjects Research Institutional Review Board. Subsequently, subjects donating blood cells consented according to the approval of the Institutional Review Board committee.

Data Availability

This analysis was performed on the initial release for the LINCS L1000 data set. Updated data from LINCS are available under accession No. GSE92742. The relevant same plate vehicle/vector control *z* scores calculated from the original LINCS L1000 data release and the R scripts to reproduce the analysis and figures presented here are available at https://github.com/vanandelinstitute/Lamin.

Statistical Analysis

The statistical methods used varied according to the requirements of each type of analysis performed and are summarized in the descriptions of each analysis below. The investigators

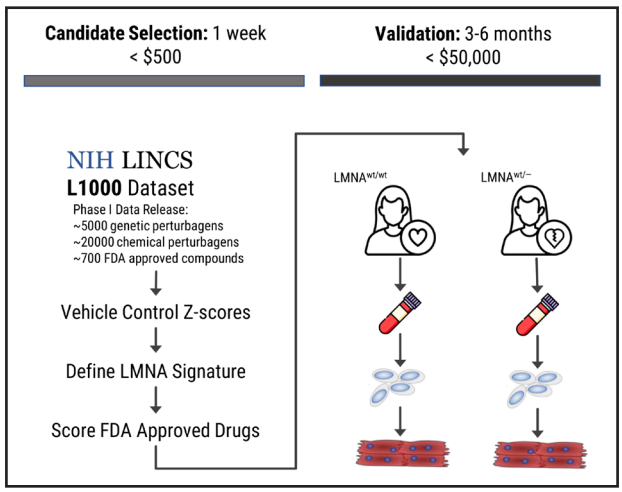


Figure 1. Overview of approach.
The Library of Integrated Network-Based Cellular Signatures (LINCS) L1000 data were downloaded, and expression levels were converted to *z* scores by comparing each treated sample with corresponding controls. From this data set, we identify a gene-specific signature (in this case, the signature describing the transcriptional consequences of *LMNA* knockdown). With this signature, gene-targeting drugs are identified. The candidate drugs are then validated, in this case by testing in patient and healthy control induced pluripotent stem cell–derived cardiomyocytes. Estimated duration and cost of this repositioning pipeline are provided for reference. FDA indicates US Food and Drug Administration.

Downloaded from http://ahajournals.org by on February 4, 2026

who performed the phenotype assessment of the cardiomyocytes (CMs) were blinded to group allocation during experiments and data collection.

LMNA Gene Signature Definition

Full details of the generation of the *LMNA* gene expression signature and drug selection based on that signature using the LINCS L1000 database are provided in the Supplemental Material (data_analysis.html). This information and the supporting data required to repeat the analysis presented here (including regenerating the figures) are freely available from github (https://github.com/vanandelinstitute/Lamin). Briefly, normalized gene expression data were obtained from the LINCS L1000 program. We then calculated the robust *z* score for each gene within each sample relative to vehicle-treated samples of the same cell type on the same 384-well plate. We extracted the *z* scores for all instances treated with short hairpins targeting *LMNA*. These data were ranked sample-wise. Second, the entire matrix of ranks (978 genes by 84 shRNA samples) was ranked, and Kolmogorov-Smirnov analysis was performed on the position of each occurrence of each gene within this vector of ranks.[17]

The resulting analysis quantifies the extent to which the expression of each gene was consistently biased up or down relative to all other genes.

Last, a bootstrapping procedure was performed to estimate the significance of the Kolmogorov-Smirnov score for each gene. We calculated Kolmogorov-Smirnov scores for 100 000 random sets of shRNA-treated samples (84 samples in each set to match the *LMNA* set) in the LINCS data. The resulting *P* values were adjusted for multiple comparisons with the method of Benjamini and Hochberg to control the false discovery rate at <5%.[18]

Drug Selection

To estimate the bias in gene expression for the genes in our *LMNA* signature within each sample in the L1000 data set treated with an FDA-approved compound, we used the XSUM metric because there is some evidence it is among the more performant algorithms for connectivity map (CMAP)–type data.[19] The XSUM limits its search to the top n variable genes. However, because the L1000 data set is already confined to the 978 most variant genes in the genome as determined by the LINCS program, we did not filter the gene set further. Therefore, we take the sum of the *z* scores for our upregulated genes and subtract the sum of the *z* scores of the downregulated genes within the 978 L1000 genes for each drug-perturbed instance. Because there are multiple instances per drug, we collapsed these scores to a single score per drug by taking the median. We again used a bootstrapping procedure to estimate the significance of the score of each drug relative to random perturbations. We scored 10 000 random gene signatures (each with the same number of upregulated and downregulated genes as the *LMNA* signature) to estimate how specific each drug was to the *LMNA* signature. Drugs were then ranked on the basis of their bootstrapped *P* value.

Generation of Human iPSCs

Peripheral blood mononuclear cells were isolated with a Ficoll-Paque PLUS gradient (GE Healthcare) and expanded as previously reported.[20] For reprogramming, 1 million peripheral blood mononuclear cells were plated in medium supplemented with 4 OSKM reprogramming factors (CytoTune-iPS Sendai Reprogramming Kit, Life Technologies) according to manufacturer recommendations. The medium was changed after 24 hours of transfection and transferred to E7N medium (E8 medium minus TGFβ1 and 200 µmol/L sodium butyrate) on day 3. Colonies were picked into 1 well of a 12-well plate (1 colony in each well) on approximately day 20 and cultured in E8 with 10 µmol/L Y-27632 (Selleckchem). hiPSCs were then expanded into 6-well plates (coated with 1:200 growth factor–reduced Matrigel) and maintained in E8 medium. Confluent hiPSCs were passaged every 4 days with 0.5 mmol/L EDTA.

Differentiation of hiPSCs to Cardiomyocytes

hiPSCs were routinely maintained in 6-well plates as described previously. Cells were grown to reach 90% confluence and then subjected to differentiation in RPMI/B27 without insulin medium (Life Technologies) supplemented with 6 µmol/L CHIR99021 (Selleckchem). After 48 hours, the cells were subjected to the same medium supplemented with 4 µmol/L IWR-1-endo (Selleckchem). On day 7, the medium was changed to RPMI-B27 with insulin and exchanged every other day. Beating hiPSC-CMs usually can be observed around day 7 to 10. On day 11, the medium was switched to RPMI-B27 without D-glucose (Life Technologies) for 4 days to purify CMs. For drug treatment and function analysis, purified iPSC-CMs were dissociated with TrypLE Express (Life Technologies) and replated to Matrigel-coated plates accordingly.

Drug Treatment

The indicated drug compounds were reconstituted from powder in dimethyl sulfoxide (DMSO) to a working concentration of 10 mmol/L. Drugs were then added to the cell culture wells to a final concentration of 10 µmol/L, and the cells were returned to the cell culture incubator for the indicated times. Controls were treated with DMSO alone.

Quantitative Real-Time Polymerase Chain Reaction

RNA was extracted with a QIAGEN RNeasy kit following manufacturer instructions. cDNA was synthesized from 100 ng total RNA with the High Capacity RNA-to-cDNA kit (ThermoFisher Scientific). Real-time polymerase chain reaction was performed with TaqMan Gene Expression Master Mix and TaqMan probes (*GAPDH*, Hs02758991_g1; *TNNT2*, Hs00165960_m1; *MYH6*, Hs01101425_m1; *MYH7*, Hs01110632_m1). Polymerase chain reactions were conducted on a 7900HT real-time polymerase chain reaction system (ThermoFisher Scientific) with triplicates and assessed with $\Delta\Delta$Ct relative quantification method normalizing to GAPDH housekeeping gene.

Western Blot

SDS-PAGE and blotting were carried out in the usual fashion. Antibodies used were mouse anti-phospho-Akt (Ser473, clone D9W9U, Cell Signaling No. 12694), mouse anti-phospho-Erk1/2 (Thr202/Tyr204, clone E10, No. 9106), polyclonal rabbit anti-Erk1/2 (No. 9102), mouse anti-phospho-Akt (Ser473,

Downloaded from http://ahajournals.org by on February 4, 2026

Clone D9W9U clone, No. 12694), and polyclonal rabbit anti-Akt antibody (No. 9272).

Field Potential Recordings

The cardiac field potentials (FPs) were recorded with the Maestro multielectrode array (MEA) platform (Axion Biosystems).[21–23] Human iPSC-CMs were enzymatically dissociated with TrypLE Select Enzyme (10×; ThermoFisher Scientific) for 5 to 6 minutes at 37 °C and directly seeded on the electrode area into 48-well MEA plate (Axion Biosystems) previously coated with Matrigel. Cardiomyocytes were maintained in medium that was replaced every 3 days. FP duration (FPD) recording were performed 14 days after seeding at 37 °C in a 5% CO_2 environment in accordance with manufacturer instructions. Baseline cardiac electrical activity was recorded on the MEA plate, and hiPSC-CMs were subsequently treated with olmesartan (10 µmol/L) or vehicle (DMSO), respectively. After 48 hours, the drug effect on FP activity was recorded. Data acquisition was performed with AxIS Navigator software, and data analysis was achieved with the Cardiac Analysis Tool (Axion Biosystems). Three primary end points were derived from the cardiac FP in the baseline and postdrug condition: FPD (milliseconds), beat period (seconds), and spike amplitude (millivolts). To account for rate-dependent effects, FPD was also reported as beat rate corrected with the Fridericia correction. Data were presented as percentage change between baseline and drug testing recordings. Drug effects of each group were recorded in ≥6 independent replicates. Arrhythmia-like events recorded by MEA were identified and categorized manually as described previously.[21] Data were presented as mean±SEM. Comparisons were conducted with 1-way ANOVA followed by the Tukey test.

Fabrication of Engineered Heart Tissues

Engineered heart tissues (EHTs) were fabricated in a 24-well plate around silicone posts in agarose casting molds with a minimum concentration of 1.5×10^6 iPSC-CMs from both a healthy control and an *LMNA* patient. For each EHT, a master mix of 100 µL total was prepared on ice containing 10 µL Matrigel (Corning), 2.5 µL of 200 mg/mL fibrinogen (to make a working concentration of 5 mg/mL), 5.5 µL of 2× high-glucose DMEM containing 10% horse serum and 1% penicillin/streptomycin, 79.4 µL high-glucose DMEM and 1% L-glutamine (ThermoFisher), 0.1 µL Y-27632 (Rock inhibitor), and 2.5 µL thrombin (Sigma). The EHT matrix was incubated for 2 hours at 37 °C and then transferred to a new 24-well plate containing RPMI/B27 medium containing 10% knockout serum. Every 2 days, medium was replaced with fresh RPMI/B27 medium containing 33 µg/mL aprotinin (Sigma).

High-Content Video-Based iPSC-CM Contractility and EHT Force Analysis

hiPSC-CMs were plated onto Matrigel-coated 96-well plates (40 000 per well) as described previously. After treatment with drugs, iPSC-CMs were examined on Sony SI8000 Live Cell Imaging System (Sony Biotechnology) with CO_2 and 37 °C incubation. Cell activities were recorded the beating video at a high frame rate (150 frames/s); focus and light conditions were automated controlled by the SI8000 software. After data acquisition, displacements and magnitudes of CM motions were calculated and presented with a motion detection algorithm by SI8000 software. Like iPSC-CMs, contractility of EHTs was recorded with high-resolution motion-capture tracking, and functional parameters were assessed from the averaged contraction-relaxation waveforms from 8- to 10-second video recordings. Moreover, the SI8000R Analyzer software was further leveraged to track movement of each EHT after drug treatments. Using a custom-written MATLAB software maximum, post deflection from rest and length signals was calculated from each contraction cycle. This deflection of the post was then converted to force with the equation for deflection of an end-loaded cantilever beam with elastic modulus of 1.7 MPa, radius of 0.5 mm, and length of 10 mm.

Treatment conditions were compared with 1-way ANOVA followed by pairwise *t* tests when ANOVA indicated significant difference between groups.

RESULTS

LMNA Gene Signature Definition

We first obtained the level 3 (normalized expression value) data from the National Institutes of Health LINCS program. For our analysis, we used only the 978 directly measured genes (the landmark genes). The *z* scores published by the LINCS program at the time we conducted this analysis were calculated for each sample compared with all the other samples on each experimental plate. For the purposes of our analysis, we wanted the *z* scores calculated against controls only. Therefore, we calculated robust *z* scores for each treated sample with $(x-\mu_{v1/2})/(MAD\times1.482)$, where $\mu_{v1/2}$ is the median value for the appropriate vehicle or empty vector controls on each plate.

We next sought to identify those genes that consistently and specifically shifted in expression because of *LMNA* knockdown (Figure 2). To find these genes that consistently responded to *LMNA* knockdown, we performed a "rank of ranks" analysis. First, we ranked the *z* scores (genes) within each sample that was treated with *LMNA*-targeting short hairpin (n=84) constructs in the LINCS data set (Figure 2A, left). Next, we combined these ranks across the samples into a single list and tested each gene to see how random their ranks were across samples by means of the Kolmogorov-Smirnov statistic, essentially inverting the traditional Gene Set Enrichment Analysis[22] to quantify the enrichment of each gene across a set of samples rather than a set of genes within each sample (Figure 2A, right).

We also wanted to know how specific these gene expression changes were to *LMNA* knockdown compared with artifactual components of the experimental setup of the LINCS assay or nonspecific cellular responses to genetic perturbation. Therefore, we repeated the aforementioned gene scoring using 100 000 random sets of gene-perturbed samples, with each set having the same number of samples as our *LMNA*-perturbed set. This procedure allowed us to determine the probability that the

Downloaded from http://ahajournals.org by on February 4, 2026

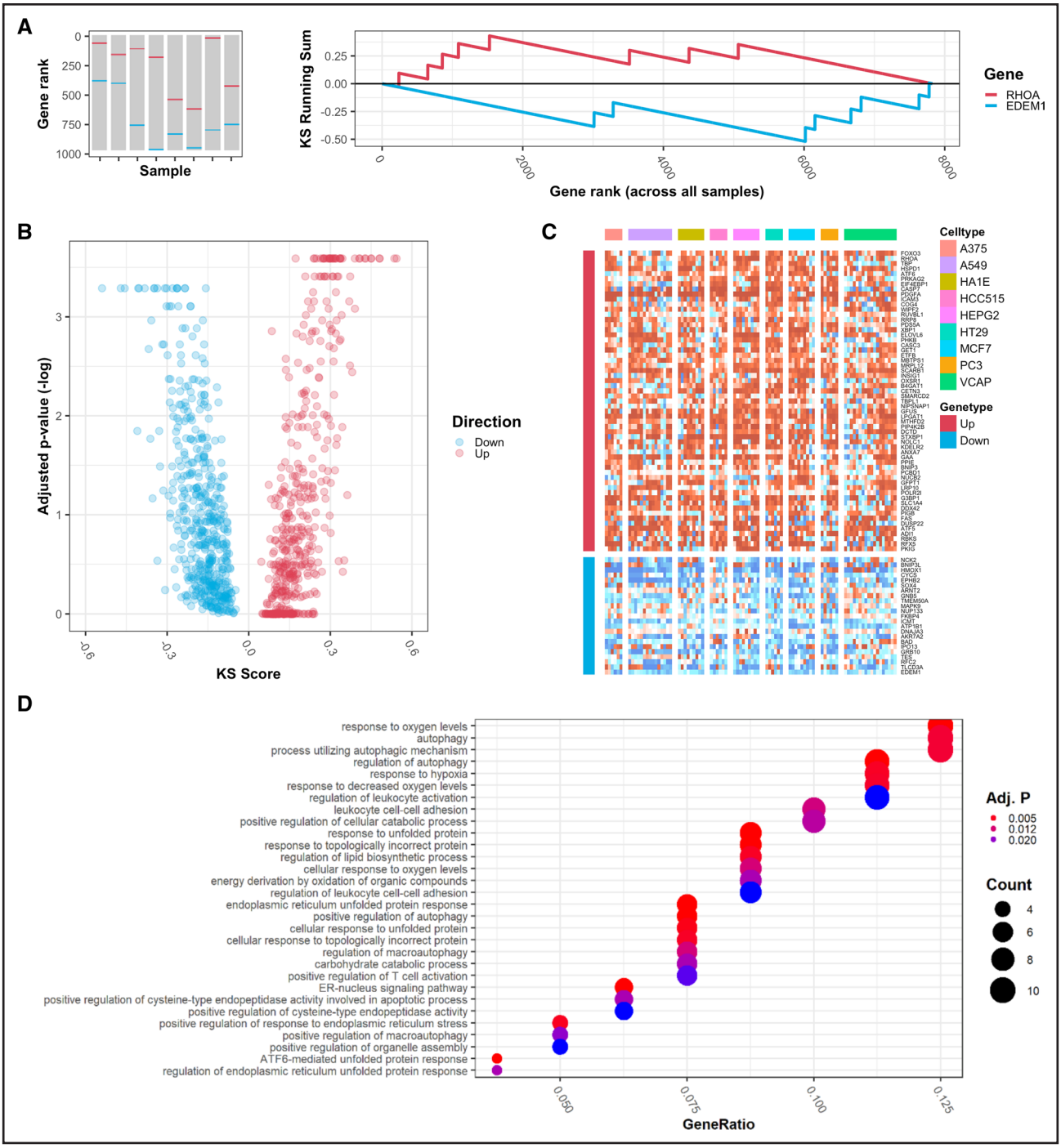


Figure 2. Establishment of *LMNA* knockdown gene signature.
A, Genes were scored according to how consistently and how strongly they were upregulated or downregulated by *LMNA* knockdown in the Library of Integrated Network-Based Cellular Signatures L1000 database. Permutation of random sets of short hairpin perturbed samples was used to estimate the significance of these expression shifts. We selected those genes with both a Kolmogorov-Smirnov (KS) score >0.3 and a −log *P*>0.3 as our *LMNA* knockdown gene set. **B**, The *LMNA* knockdown gene signature showed fairly uniform upregulation or downregulation across all samples in the L1000 data set across all cell types treated. **C**, Gene Ontology enrichment analysis suggests that the genes in this signature are involved in pathways related to autophagy and apoptosis. **D**, Gene Ontology terms–related *P* value and gene ratio and count.

differential expression statistic calculated for each gene from the *LMNA*-perturbed samples was a unique feature of *LMNA* perturbation as opposed to a generic response frequently observed in random sets of gene-perturbed samples (Figure 2B). The resulting bootstrapped *P* values were adjusted to control the false discovery rate.[18]

Although the most extreme differential scores were all highly significant as determined by this analysis, several

Downloaded from http://ahajournals.org by on February 4, 2026

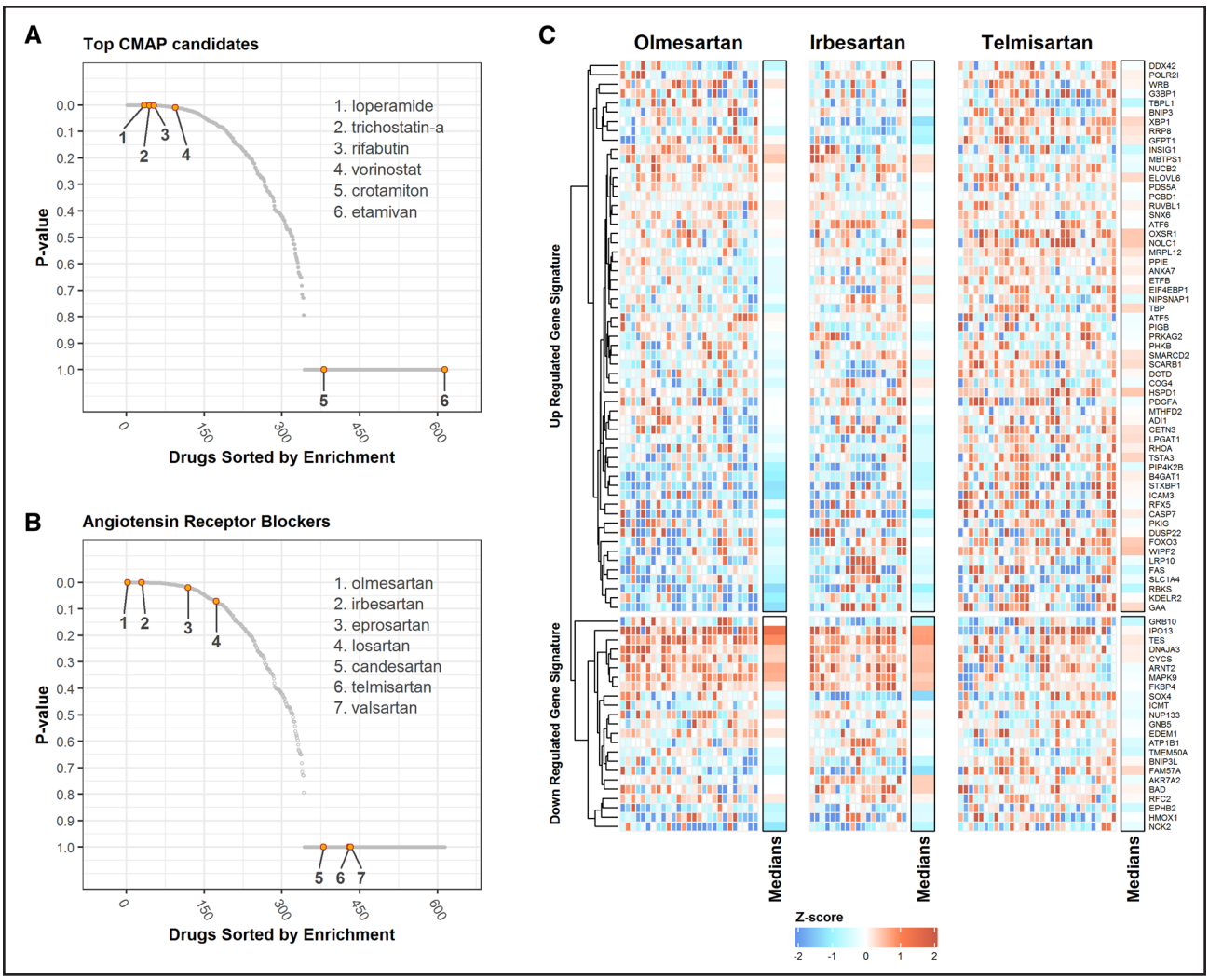


Figure 3. The *LMNA* knockdown signature was used to identify drugs that could reverse these transcriptional changes in vitro. **A**, We first submitted the gene signature to the original connectivity map (CMAP) web tool. Of the top 6 drugs that most strongly reversed the *LMNA* knockdown signature according to that tool, 4 also exhibited highly significant reversal of this signature in the Library of Integrated Network-Based Cellular Signatures L1000 data, suggesting that there was some coherence between the various platforms and analytical approaches used by these systems. **B**, We note that 2 angiotensin receptor blockers were among the most highly significant drugs based on their ability to reverse the *LMNA* knockdown signature. Indeed, 3 of 7 angiotensin receptor blockers (ARBs) in the database showed gene set enrichment with a false discovery rate–adjusted $P<0.05$. **C**, These ARBs do not reverse the entirety of the *LMNA* signature. Rather, high-scoring ARBs (olmesartan and irbesartan) reverse highly overlapping segments of the *LMNA* signature, whereas telmisartan (a low-scoring drug) exhibits a more random expression pattern for the genes in the *LMNA* knockdown signature.

genes had intermediate differential expression scores that were nevertheless highly significant compared with random perturbations (Figure 2B). We selected those genes with false discovery rate–adjusted $P<0.001$ and an enrichment (Kolmogorov-Smirnov) score >0.2 (absolute value) as our *LMNA* knockdown signature (Table S1). These genes were quite consistently downregulated or upregulated across samples and cell lines (Figure 2C).

PDGFA was among the most upregulated genes in our *LMNA* knockdown signature, consistent with recent work documenting activation of PDGF signaling in CMs harboring a mutation of *LMNA*.[12] In addition, we performed Gene Ontology term enrichment analysis[21] on the Biological Function Ontology for the genes in our signature (Figure 2C). This analysis identified multiple Gene Ontology terms related to autophagy, apoptosis, and hypoxia response that were enriched in our *LMNA* knockdown gene set, also consistent with previous work (Figure 2D).[7,22]

Candidate Drug Identification

We next sought to identify drugs that could reverse this *LMNA* knockdown signature. There is some evidence that parametric approaches are quantifiably superior to nonparametric scoring metrics for this type of gene signature enrichment analysis. As a result, we chose the XSUM statistic to quantify reversal of the *LMNA*

Downloaded from http://ahajournals.org by on February 4, 2026

signature within drug-treated samples.[23] Because there were many samples per drug in the L1000 data set, we took the median score for each drug. In this way, we informatically screened >600 FDA-approved drugs for their ability to reverse the *LMNA* knockdown signature. We again assigned *P* values to these scores by permutation (based on 10 000 random gene signatures).

As validation, we submitted our *LMNA* signature to the original CMAP enrichment tool (https://www.broadinstitute.org/connectivity-map-cmap)[24] to see what drugs could reverse our knockdown signature as determined by the CMAP connectivity score (Table S2). Of the FDA-approved drugs in our L1000 data set, 6 demonstrated a significant ability to reverse the *LMNA* signature according the CMAP enrichment tool. Of these 6 drugs, 4 also had significant enrichment scores based on the XSUM statistic derived from the L1000 data set (Figure 3A). These 4 drugs had stronger enrichment scores based on the original CMAP tool compared with the 2 drugs that were not enriched in the L1000 analysis.

We noted that multiple angiotensin receptor blockers exhibited significant reversal of the *LMNA* knockdown signature (Table S3; Figure 4B), including olmesartan, which was among the group of drugs that reversed this

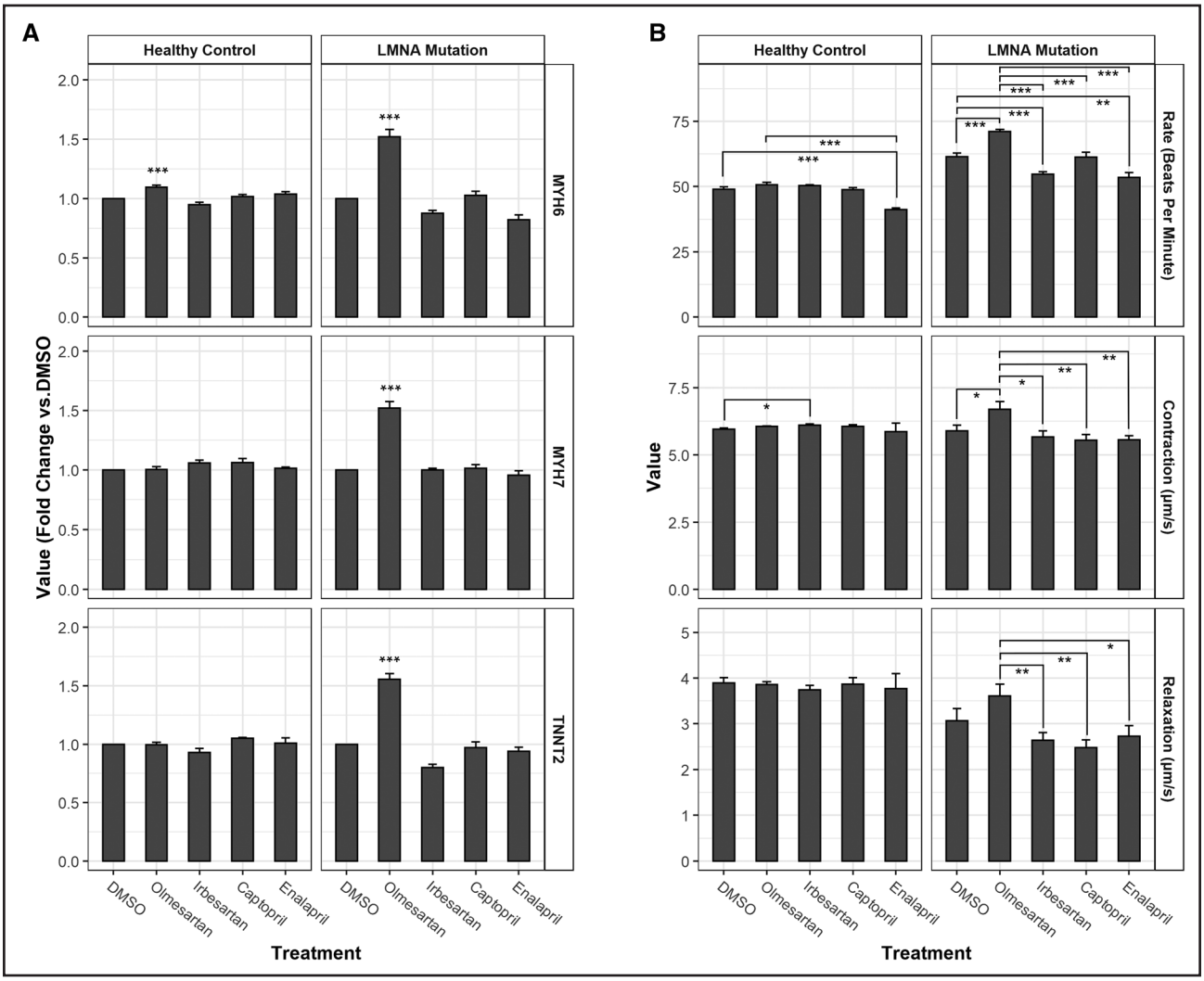


Figure 4. In vitro validation of the effect of predicted drugs on CM transcription and phenotype.
A, We measured the expression of 3 cardiac markers (*MYH6*, *MYH7*, and *TNNT2*) by quantitative polymerase chain reaction in cardiomyocytes (CMs) derived from induced pluripotent stem cells from a healthy control and a patient harboring an *LMNA* mutation. Fold change relative to dimethyl sulfoxide (DMSO)–treated controls was calculated for each marker after 48 hours of treatment with the drugs shown, which include 2 of the angiotensin receptor blockers (ARBs) identified by our analysis and 2 angiotensin-converting enzyme inhibitors used as controls because of their canonical activity on the same pathway as ARBs. Significant increases from DMSO-treated samples are indicated (***$P \leq 0.001$ by 2-tailed *t* tests). **B**, Beating rate, contractile velocity, and relaxation velocity were measured by quantitative video microscopy in both control and *LMNA*-mutant cardiomyocytes. Rate is expressed in beats per minute. Contractile force is expressed in micrometers per second. Significant differences from DMSO- and olmesartan-treated samples are indicated (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$ by 2-tailed *t* tests; where multiple significant comparisons are shown, preceding 1-way ANOVA analysis demonstrated $P < 0.01$ in each case). All results are from duplicate experiments, with 3 technical replicates per experiment (total n=6). All error bars are mean+SEM.

Downloaded from http://ahajournals.org by on February 4, 2026

signature most significantly. However, not all angiotensin receptor blockers exhibited this feature. It has previously been demonstrated that the various members of this drug class have variable transcriptional effects.[25–27] When we analyzed all angiotensin receptor blockers in our data set as a group, this class of drugs collectively exhibited a significant reversal of the *LMNA* knockdown signature, but this effect was weaker for the entire class than for olmesartan or irbesartan specifically (false discovery rate, 0.0147 versus 0.0001 and 0.0021, respectively).

When we examined the effect of the highest-scoring angiotensin receptor blockers (olmesartan and irbesartan) on the expression of the individual genes in the *LMNA* knockdown signature, we noted that these drugs downregulated many of the genes upregulated by *LMNA* knockdown and vice versa (Figure 4C). No such trend was apparent for telmisartan, which did not exhibit significant enrichment in our analysis. However, the transcriptional effects of olmesartan and irbesartan were not universal with respect to the *LMNA* regulated genes, indicating that these drugs were acting on only portions of the *LMNA* knockdown "axis."

In Vitro Validation

We next tested whether treatment with olmesartan, irbesartan, or both could influence the expression of cardiac markers or function of CMs derived from iPSCs harboring a disease-related *LMNA* nonsense mutation. These cells harbor a heterozygous insertion of a guanine between nucleotides 348 and 349, causing a frameshift mutation at codon 117 and resulting in a premature stop at codon 129. In iPSC-CMs from a healthy control patient treated for 48 hours, a slight increase in *MYH6* was observed in olmesartan-treated cells, and a slight increase in *TNNT2* expression was observed in captopril-treated cells (Figure 4A). In contrast, there was a 1.5-fold increase in *MYH6*, *MYH7*, and *TNNT2* in olmesartan-treated iPSC-CMs derived from a patient harboring a nonsense *LMNA* mutation. Other treatments tested either had no effect on these markers or suppressed their expression. We hypothesized that increased expression of these sarcomeric genes would correspond to improved contractility in these *LMNA*-mutant cells.

To test this hypothesis, we examined the effects of these drugs on the functional properties of these cells. For this, we examined the contractile properties of healthy control and *LMNA* iPSC-CMs. Irbesartan treatment of iPSC-CMs derived from a control patient produced a very small (but statistically significant) increase in contraction velocity (Figure 4B) but no significant effect on contractile rate relative to DMSO-treated controls, whereas olmesartan had no effect on the behavior of these cells relative to the DMSO control. In contrast, olmesartan treatment of iPSC-CMs derived from the patient with an *LMNA* mutation resulted in a marked increase

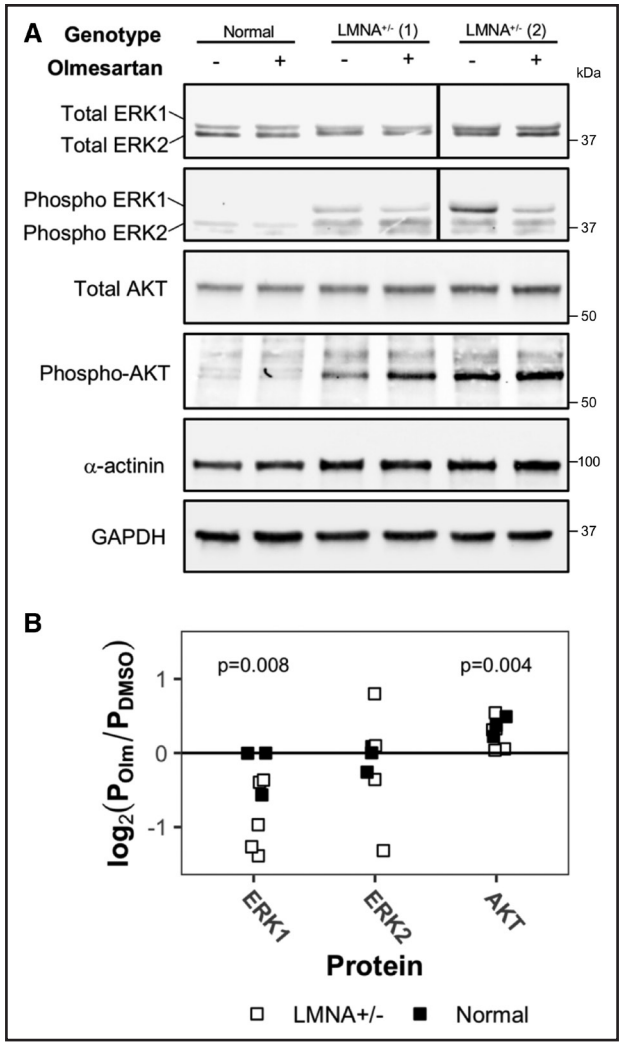


Figure 5. Effect of olmesartan treatment on elements of the PDGF signaling pathway.
A, Cardiomyocytes differentiated from induced pluripotent stem cells from a healthy donor (normal) and 2 parallel differentiations from patient cells heterozygous for the *LMNA* c.348_349insG mutation (*LMNA*$^{+/-}$) were maintained in culture and treated with olmesartan (+) or dimethyl sulfoxide (DMSO; —). Protein was extracted after 48 hours of treatment and subjected to SDS-PAGE and blotting for the indicated proteins. Loading controls were α-actinin (for extracellular signal–regulated kinase [ERK]) or GAPDH (for AKT). Black lines in top 2 panels indicate where an empty lane was removed from the image. **B**, Quantification by densitometry. Log ratio of phosphorylated to total protein is shown. *P* values are for *t* test vs the null hypothesis that the log ratio=0. PDGF indicates platelet-derived growth factor.

in both contraction velocity and beating rate. Although there was a trend toward increased relaxation velocity in olmesartan-treated cells compared with DMSO controls, this difference was not significant. However, olmesartan treatment was associated with significantly increased relaxation velocity compared with the other drugs tested. These observations support the hypothesis that olmesartan will not exacerbate any underlying diastolic dysfunction and may in fact be favorable to other heart failure medications with respect to diastolic function.

Downloaded from http://ahajournals.org by on February 4, 2026

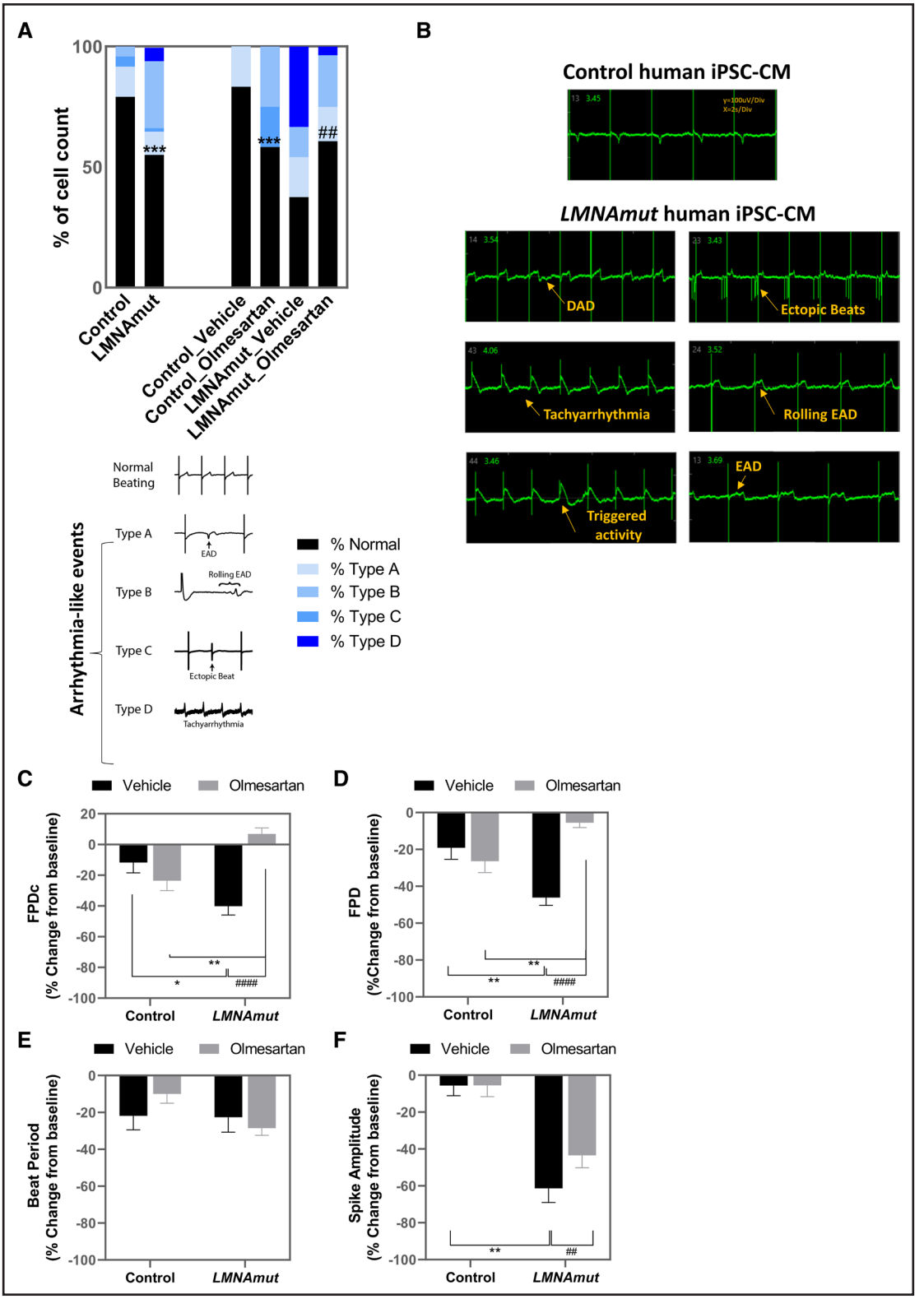


Figure 6. Olmesartan ameliorates the arrhythmic phenotype in *LMNA*-mutant iPSC-CMs.
A, Pooled percentage of arrhythmia-like events in *LMNAmut* and control human induced pluripotent stem cell–induced cardiomyocytes (hiPSC-CMs) before and after 48 hours of olmesartan or vehicle (dimethyl sulfoxide [DMSO]) treatment detected with a Maestro multielectrode array and classified according representative traces of 4 different cellular arrhythmogenic events (**bottom**) from Axion Biosystems. All data are represented as percentages, and comparisons were conducted with the Fisher exact test. ***$P<0.001$ vs control iPSC-CMs. ##$P<0.01$ vs *LMNAmut* iPSC-CMs. **B**, Representative field potential (FP) recording waveforms from control and *LMNAmut* hiPSC-CMs. Differences in signal amplitude and examples of arrhythmia-like events found in *LMNAmut* hiPSC-CMs (yellow). Percentage of change from baseline of Fridericia (*Continued*)

Downloaded from http://ahajournals.org by on February 4, 2026

Figure 6 Continued. rate–corrected FP duration (FPDc; **C**) measured in milliseconds, FP duration (FPD; **D**) measured in milliseconds, beat period measured in seconds (**E**), and depolarization spike amplitude measured in millivolts (**F**). Data were collected from ≥6 independent measurements per each group. All data are represented as mean±SEM, and comparisons were conducted with 1-way ANOVA followed by the Tukey test. **P*<0.05, ***P*<0.01 vs control iPSC-CMs. ##*P*<0.01, ####*P*<0.01 vs *LMNAmut* iPSC-CMs.

Mechanism of Action

It has recently been shown that upregulation of PDGF signaling is responsible for aspects of cardiomyopathy associated with *LMNA* haploinsufficiency. Consistent with these findings, we observed that *PDGFA* was one of the most strongly and consistently upregulated genes in cells treated with *LMNA* targeting short hairpins in the LINCs database (Figure 2C, arrowhead). To determine whether olmesartan affects PDGF signaling, we performed Western blotting for several mediators of PDGF signaling in iPSC-CMs (Figure 5). Although total extracellular signal–regulated kinase (ERK) was similar between normal and *LMNA*-haploinsufficient CMs, a substantial increase in phosphorylated ERK1 was observed in the *LMNA* iPSC-CMs. The increase in phosphorylated ERK1 was reversed by olmesartan treatment of these cells. Phosphorylated ERK1 as a proportion of total ERK1 was significantly decreased by olmesartan treatment over multiple biological (n=2) and technical (n=3) replicates in both healthy control and *LMNA* iPSC-CMs, although the strongest effects were observed among *LMNA*-haploinsufficient CMs. No such significant difference was observed in ERK2 phosphorylation. In contrast, Akt phosphorylation was modestly but significantly decreased by olmesartan treatment in both healthy and *LMNA*-haploinsufficient cells.

Olmesartan Ameliorates the Arrhythmogenic Phenotype in *LMNA* iPSC-CMs

To understand how olmesartan affects the electrophysiological properties of iPSC-CMs, we next examined the cardiac electrical activity at the monolayer level using MEAs. Our previous study demonstrated that patient-specific iPSC-CMs recapitulated the disease arrhythmogenic phenotype of patients with *LMNA*-related dilated cardiomyopathy.[12] The electrophysiological studies at the single-cell level showed that *LMNA* iPSC-CMs exhibited an increased arrhythmogenicity compared with healthy controls, which is a risk factor predisposing to arrhythmias. We evaluated the arrhythmic susceptibility of *LMNA*-mutant iPSC-CMs under each treatment condition. Our results demonstrated that *LMNA* mutation induces a substantial increase in the number of arrhythmia-like events compared with the control group (Figure 6A), which is consistent with the study described previously.[12] Four types of arrhythmia-like events were identified with MEA and classified manually according the representative traces from Axion Biosystems shown in Figure 6A (bottom) and previously described.[28] Findings showed that olmesartan decreased the frequency of arrhythmia-like events on *LMNA* iPSC-CMs compared with vehicle-treated *LMNA* iPSC-CMs (Figure 6A). In addition, *LMNA* iPSC-CMs exhibited arrhythmias such as delayed afterdepolarizations, triggered activity, and early after depolarization events that were more severe than those found in control iPSC-CMs, and this was ameliorated by olmesartan treatment (Figure 6B).

We found that *LMNA* mutation significantly shortens the FPD and corrected FPD of iPSC-CMs compared with healthy controls (Table S4; Figure 6C and 6D), which is consistent with the action potential duration data found for this patient-specific iPSC-CM cell line in our previous study.[12] No changes were found in the beat period (Table S4; Figure 6E). Furthermore, *LMNA* mutation strongly reduces the amplitude of cardiac depolarization spike (Table S4 and Figure 6F), which is related to conduction abnormalities. Olmesartan completely reverses the effect of *LMNA* mutation on FPD and corrected FPD (Figure 6C and 6D) and partially reverses the effect of mutation on depolarization spike amplitude (Figure 6F) after 48 hours. Olmesartan did not induce change in the beating period in control and *LMNA* iPSC-CMs (Figure 6E). Olmesartan had no significant effect on the electrophysiological properties of control iPSC-CMs. These findings suggest that olmesartan ameliorates the arrhythmogenic phenotype in *LMNA* iPSC-CMs, suggesting a therapeutic strategy for *LMNA*-related dilated cardiomyopathy.

Olmesartan Improves Contractile Function of *LMNA*-Mutant EHTs

Last, we wanted to know whether olmesartan treatment could, in addition to ameliorating the arrhythmogenicity of iPSC-CMS, improve the contractile behavior of these cells. This is particularly relevant in view of the fact that patients with *LMNA* mutation experience not only arrhythmias but also heart failure. We created EHTs composed of iPSC-CMs in a Matrigel matrix adhered to silicon posts, thus enabling us to quantify contractile force through deflection of the posts (Figure 7A and 7B). Consistent with our 2-dimensional iPSC-CM results described previously, we again observed that drug treatments had no effect on the beating rate, contraction velocity, or relaxation velocity in healthy control EHTs. However, in contrast, a significant increase was seen in both contraction and relaxation velocity in *LMNA* EHTs treated with olmesartan relative to both vehicle control (DMSO) and other drug

Downloaded from http://ahajournals.org by on February 4, 2026

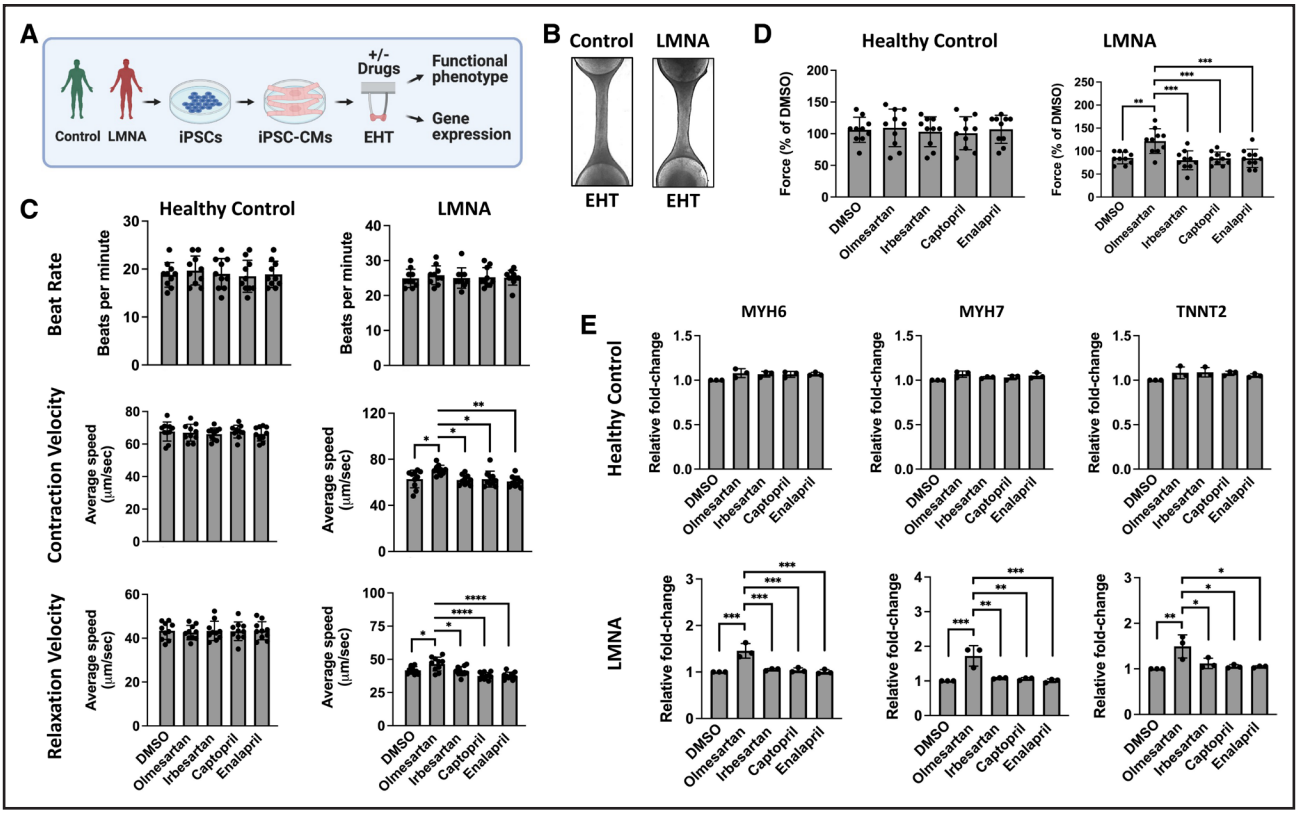


Figure 7. Olmesartan improves function in *LMNA*-mutant EHTs.
A, Schematic diagram of experimental setup for engineered heart tissue (EHT) experiments. **B**, Representative micrographs of engineered tissue adhered to silicon posts to enable measurement of contractile force through post deflection. **C**, Drug treatment had no effect on beating rate, contraction velocity, or relaxation velocity in wild type EHTs. However, there was a significant increase in both contraction and relaxation velocity in *LMNA*-mutant EHTs treated with olmesartan relative to both vehicle control (dimethyl sulfoxide [DMSO]) and other drug treatments. **D**, Similarly, olmesartan treatment produced an increase in EHT contractile force relative to DMSO and other drug treatment in *LMNA*-mutant EHTs but not wild-type EHTs. **E**, These changes in contractile velocity and force were accompanied by an increase in expression of key sarcomeric genes in olmesartan-treated *LMNA*-mutant EHTs, whereas no such effect was observed in wild-type EHTs. **P*<0.05, ***P*<0.01, ****P*<0.001, *****P*<0.0001 by the Tukey honestly significant difference. When significant results are reported, the *P* value for 2-way ANOVA of the corresponding data set was <0.01 in each case.

treatments. Treatment with the other studied drugs had no such effect (Figure 7C). Along with increased contraction velocity, we observed an increase in contractile force after olmesartan treatment as quantified by silicon post displacement by the EHTs. This effect was observed only in *LMNA* EHTs treated with olmesartan. Treatment with other drugs or treatment of control EHTs had no significant effect (Figure 7D). As with our 2-dimensional experiments, these changes in contractile velocity and force were associated with changes in expression of sarcomeric genes. Specifically, *MYH6*, *MYH7*, and *TNNT2* increased in expression in *LMNA* EHTs treated with olmesartan. This effect was not observed after treatment with the other drugs studied or when wild-type EHTs were treated with any of these compounds.

DISCUSSION

With advances in the treatment of coronary artery disease and cardiac,[29] mortality resulting from heart disease is declining, whereas the prevalence of heart failure and associated morbidities is rising.[30–32] Although ischemia and infarction are major causes of heart failure, genetic mutations may contribute to the pathology either through interaction with the development of or response to combination with coronary artery disease[33] or independently of it.

Dozens of genes are known to contribute to cardiomyopathy and heart failure, with the most common including mutations in the genes encoding titin, lamin A/C, and components of the contractile apparatus such as myosin and troponin.[34] Although heart failure associated with any one of these mutations accounts for only a small proportion of the total burden of disease (typically far less than 5% for any given mutation), collectively, the burden of heart failure from genetic mutation is substantial.[35] For example, approximately one-third of all cases of dilated cardiomyopathy can be attributed to known gene mutations[36], and this number will inevitably grow as studies into the causes of heart failure progress.

Despite the substantial burden of monogenic heart disease, therapeutic progress is hampered by the small

Downloaded from http://ahajournals.org by on February 4, 2026

number of patients affected by any given mutation. For this reason, high-efficiency approaches to drug screening and development are needed if therapies targeting specific genetic causes of heart disease are to be established. Here, we describe such an approach based on in silico screening for new applications of existing FDA-approved compounds. This screening identified candidate therapies for heart failure attributable to *LMNA* haploinsufficiency. One of these candidates, olmesartan, reversed the effects of *LMNA* haploinsufficiency in vitro in terms of expression of sarcomeric genes and both contractile function and arrhythmogenicity of iPSC-CMs. Supporting our findings, several studies have demonstrated previously that olmesartan treatment could alleviate cardiac hypertrophy and consequently reduce the severity of arrhythmias in different cardiomyopathy animal models.[37–40] In addition, this analysis points to a novel function of olmesartan: inhibition of the PDGF axis by means of decreased phosphorylation of ERK1.

This analysis suggests a novel application of olmesartan specifically for patients with heart failure related to *LMNA* haploinsufficiency while suggesting a possible role for olmesartan in other diseases that are the result of derangements of PDGF signaling. Such diseases include neurodegeneration and cancer,[41] diseases that often coincide in patients with heart disease, suggesting the opportunity for synergistic adjuvant therapeutic strategies in these patients.

Mitogen-activated protein kinase/ERK signaling has been strongly implicated in the literature as playing a key role in heart disease in general[42] and *LMNA* mutation–associated cardiomyopathy in particular.[14] Some authors of this report have recently shown that upregulated PDGF signaling plays a central role in aspects of *LMNA* mutation–associated dysfunction of CMs.[12] Additional evidence indicates that activation of ERK1/2, a downstream regulator of PDGF signaling, is both associated with and necessary for CM pathology associated with *LMNA* mutation.[43,44]

Conclusions

Our results demonstrate that the LINCS L1000 database can be exploited to perform an in silico screen for repositionable drugs that reverse the transcriptional consequences of specific genetic perturbations. One of 2 treatments we identified in such a way for *LMNA* mutation–related cardiomyopathy showed a favorable response in our in vitro model, suggesting that this approach to drug repositioning for rare diseases may be a promising alternative to the traditional drug development pipeline.

ARTICLE INFORMATION

Received December 14, 2021; accepted February 17, 2025.

Affiliations

DeVos Cardiovascular Research Program, Fredrik Meijer Heart and Vascular Institute, Spectrum Health and Van Andel Institute, Grand Rapids, MI (E.J.K., J.F., E.E., S.J.). Department of Pediatrics and Human Development, Michigan State University, Grand Rapids (E.J.K.). Helen DeVos Children's Hospital, Corewell Health, Grand Rapids, MI (E.J.K.). Cardiovascular Institute (N.S., C.L., G.M.-P., S.M.W., J.C.W., S.J.), Division of Vascular Surgery, Department of Surgery (N.S.), Department of Medicine, Division of Cardiovascular Medicine (S.M.W., J.C.W.), and Department of Radiology, Stanford University, CA (J.C.W.). Skåne University Hospital, Lund University, Lund, Sweden (S.J.).

Sources of Funding

This work was funded mainly through the Richard and Helen DeVos Foundation (Dr Jovinge), Swedish Heart and Lung Foundation (Dr Jovinge), and National Institutes of Health (R01 HL141371 and R01 HL163680 to Dr Wu; R01 HL158641 and R01 HL161002 to Dr Sayed).

Disclosures

None.

Supplemental Material

Expanded Methods

Figure S1

Tables S1–S4

REFERENCES

1. Peters S, Lowy P. How the Tufts Center for the Study of Drug Development pegged the cost of a new drug at $2.6 billion. Accessed April 6, 2025. https://cost_study_backgrounder.pdf
2. Subramanian A, Narayan R, Corsello SM, Peck DD, Natoli TE, Lu X, Gould J, Davis JF, Tubelli AA, Asiedu JK, et al. A next generation connectivity map: L1000 platform and the first 1,000,000 profiles. *Cell*. 2017;171:1437–1452.e17. doi: 10.1016/j.cell.2017.10.049
3. Stegmaier K, Ross KN, Colavito SA, O'Malley S, Stockwell BR, Golub TR. Gene expression-based high-throughput screening (GE-HTS) and application to leukemia differentiation. *Nat Genet*. 2004;36:257–263. doi: 10.1038/ng1305
4. Stegmaier K, Corsello SM, Ross KN, Wong JS, Deangelo DJ, Golub TR. Gefitinib induces myeloid differentiation of acute myeloid leukemia. *Blood*. 2005;106:2841–2848. doi: 10.1182/blood-2005-02-0488
5. Fatkin D, MacRae C, Sasaki T, Wolff MR, Porcu M, Frenneaux M, Atherton J, Vidaillet HJ Jr, Spudich S, De Girolami U, et al. Missense mutations in the rod domain of the lamin A/C gene as causes of dilated cardiomyopathy and conduction-system disease. *N Engl J Med*. 1999;341:1715–1724. doi: 10.1056/NEJM199912023412302
6. Pan H, Richards AA, Zhu X, Joglar JA, Yin HL, Garg V. A novel mutation in LAMIN A/C is associated with isolated early-onset atrial fibrillation and progressive atrioventricular block followed by cardiomyopathy and sudden cardiac death. *Heart Rhythm*. 2009;6:707–710. doi: 10.1016/j.hrthm.2009.01.037
7. Siu CW, Lee YK, Ho JC, Lai WH, Chan YC, Ng KM, Wong LY, Au KW, Lau YM, Zhang J, et al. Modeling of lamin A/C mutation premature cardiac aging using patient-specific induced pluripotent stem cells. *Aging (Albany NY)*. 2012;4:803–822. doi: 10.18632/aging.100503
8. Ahmed MS, Ikram S, Bibi N, Mir A. Hutchinson-Gilford progeria syndrome: a premature aging disease. *Mol Neurobiol*. 2018;55:4417–4427. doi: 10.1007/s12035-017-0610-7
9. Butin-Israeli V, Adam SA, Goldman AE, Goldman RD. Nuclear lamin functions and disease. *Trends Genet*. 2012;28:464–471. doi: 10.1016/j.tig.2012.06.001
10. Perez-Serra A, Toro R, Campuzano O, Sarquella-Brugada G, Berne P, Iglesias A, Mangas A, Brugada J, Brugada R. A novel mutation in lamin a/c causing familial dilated cardiomyopathy associated with sudden cardiac death. *J Card Fail*. 2015;21:217–225. doi: 10.1016/j.cardfail.2014.12.003
11. Nishiuchi S, Makiyama T, Aiba T, Nakajima K, Hirose S, Kohjitani H, Yamamoto Y, Harita T, Hayano M, Wuriyanghai Y, et al. Gene-based risk stratification for cardiac disorders in *LMNA* mutation carriers. *Circ Cardiovasc Genet*. 2017;10:e001603. doi: 10.1161/CIRCGENETICS.116.001603
12. Lee J, Termglinchan V, Diecke S, Itzhaki I, Lam CK, Garg P, Lau E, Greenhaw M, Seeger T, Wu H, et al. Activation of PDGF pathway links *LMNA* mutation to dilated cardiomyopathy. *Nature*. 2019;572:335–340. doi: 10.1038/s41586-019-1406-x
13. Choi JC, Muchir A, Wu W, Iwata S, Homma S, Morrow JP, Worman HJ. Temsirolimus activates autophagy and ameliorates cardiomyopathy caused

Downloaded from http://ahajournals.org by on February 4, 2026

by lamin A/C gene mutation. *Sci Transl Med*. 2012;4:144ra102. doi: 10.1126/scitranslmed.3003875
14. Muchir A, Pavlidis P, Decostre V, Herron AJ, Arimura T, Bonne G, Worman HJ. Activation of MAPK pathways links *LMNA* mutations to cardiomyopathy in Emery-Dreifuss muscular dystrophy. *J Clin Invest*. 2007;117:1282–1293. doi: 10.1172/JCI29042
15. Galiova G, Bartova E, Raska I, Krejci J, Kozubek S. Chromatin changes induced by lamin A/C deficiency and the histone deacetylase inhibitor trichostatin A. *Eur J Cell Biol*. 2008;87:291–303. doi: 10.1016/j.ejcb.2008.01.013
16. Mattioli E, Andrenacci D, Mai A, Valente S, Robijns J, De Vos WH, Capanni C, Lattanzi G. Statins and histone deacetylase inhibitors affect lamin A/C - histone deacetylase 2 interaction in human cells. *Front Cell Dev Biol*. 2019;7:6. doi: 10.3389/fcell.2019.00006
17. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, Paulovich A, Pomeroy SL, Golub TR, Lander ES, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102:15545–15550. doi: 10.1073/pnas.0506580102
18. Patel D, Stohlman J, Dang Q, Strauss DG, Blinova K. Assessment of proarrhythmic potential of drugs in optogenetically paced induced pluripotent stem cell-derived cardiomyocytes. *Toxicol Sci*. 2019;170:167–179. doi: 10.1093/toxsci/kfz076
19. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J Roy Statist Soc Ser B*. 1995;57:289–300. doi: 10.1111/j.2517-6161.1995.tb02031.x
20. Cheng J, Xie Q, Kumar V, Hurle M, Freudenberg JM, Yang L, Agarwal P. Evaluation of analytical methods for connectivity map data. *Pac Symp Biocomput*. 2013:5–16. doi: 10.1142/9789814447973_0002
21. Burridge PW, Matsa E, Shukla P, Lin ZC, Churko JM, Ebert AD, Lan F, Diecke S, Huber B, Mordwinkin NM, et al. Chemically defined generation of human cardiomyocytes. *Nat Methods*. 2014;11:855–860. doi: 10.1038/nmeth.2999
22. Blinova K, Dang Q, Millard D, Smith G, Pierson J, Guo L, Brock M, Lu HR, Kraushaar U, Zeng H, et al. International multisite study of human-induced pluripotent stem cell-derived cardiomyocytes for drug proarrhythmic potential assessment. *Cell Rep*. 2018;24:3582–3592. doi: 10.1016/j.celrep.2018.08.079
23. Yamazaki D, Kitaguchi T, Ishimura M, Taniguchi T, Yamanishi A, Saji D, Takahashi E, Oguchi M, Moriyama Y, Maeda S, et al. Proarrhythmia risk prediction using human induced pluripotent stem cell-derived cardiomyocytes. *J Pharmacol Sci*. 2018;136:249–256. doi: 10.1016/j.jphs.2018.02.005
24. Lamb J, Crawford ED, Peck D, Modell JW, Blat IC, Wrobel MJ, Lerner J, Brunet JP, Subramanian A, Ross KN, et al. The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease. *Science*. 2006;313:1929–1935. doi: 10.1126/science.1132939
25. Benson SC, Pershadsingh HA, Ho CI, Chittiboyina A, Desai P, Pravenec M, Qi N, Wang J, Avery MA, Kurtz TW. Identification of telmisartan as a unique angiotensin II receptor antagonist with selective PPARgamma-modulating activity. *Hypertension*. 2004;43:993–1002. doi: 10.1161/01.HYP.0000123072.34629.57
26. Yuen CY, Wong WT, Tian XY, Wong SL, Lau CW, Yu J, Tomlinson B, Yao X, Huang Y. Telmisartan inhibits vasoconstriction via PPARgamma-dependent expression and activation of endothelial nitric oxide synthase. *Cardiovasc Res*. 2011;90:122–129. doi: 10.1093/cvr/cvq392
27. Kurtz TW, Pravenec M. Molecule-specific effects of angiotensin II-receptor blockers independent of the renin-angiotensin system. *Am J Hypertens*. 2008;21:852–859. doi: 10.1038/ajh.2008.202
28. Nabel EG, Braunwald E. A tale of coronary artery disease and myocardial infarction. *N Engl J Med*. 2012;366:54–63. doi: 10.1056/NEJMra1112570
29. Flaherty JD, Bax JJ, De Luca L, Rossi JS, Davidson CJ, Filippatos G, Liu PP, Konstam MA, Greenberg B, Mehra MR, et al; Acute Heart Failure Syndromes International Working Group. Acute heart failure syndromes in patients with coronary artery disease early assessment and treatment. *J Am Coll Cardiol*. 2009;53:254–263. doi: 10.1016/j.jacc.2008.08.072
30. Braunwald E. Heart failure. *JACC Heart Fail*. 2013;1:1–20. doi: 10.1016/j.jchf.2012.10.002
31. Bahit MC, Kochar A, Granger CB. Post-myocardial infarction heart failure. *JACC Heart Fail*. 2018;6:179–186. doi: 10.1016/j.jchf.2017.09.015
32. Wang Q. Molecular genetics of coronary artery disease. *Curr Opin Cardiol*. 2005;20:182–188. doi: 10.1097/01.hco.0000160373.77190.f1
33. Morita H, Seidman J, Seidman CE. Genetic causes of human heart failure. *J Clin Invest*. 2005;115:518–526. doi: 10.1172/JCI24351
34. Yang J, Xu WW, Hu SJ. Heart failure: advanced development in genetics and epigenetics. *Biomed Res Int*. 2015;2015:352734. doi: 10.1155/2015/352734
35. Mestroni L, Brun F, Spezzacatene A, Sinagra G, Taylor MR. Genetic causes of dilated cardiomyopathy. *Prog Pediatr Cardiol*. 2014;37:13–18. doi: 10.1016/j.ppedcard.2014.10.003
36. Matsushita N, Kashihara T, Shimojo H, Suzuki S, Nakada T, Takeishi Y, Mende U, Taira E, Yamada M, Sanbe A, et al. Cardiac overexpression of constitutively active Galpha q causes angiotensin II type1 receptor activation, leading to progressive heart failure and ventricular arrhythmias in transgenic mice. *PLoS One*. 2014;9:e106354. doi: 10.1371/journal.pone.0106354
37. Sukumaran V, Veeraveedu PT, Lakshmanan AP, Gurusamy N, Yamaguchi K, Ma M, Suzuki K, Kodama M, Watanabe K. Olmesartan medoxomil treatment potently improves cardiac myosin-induced dilated cardiomyopathy via the modulation of ACE-2 and ANG 1-7 mas receptor. *Free Radic Res*. 2012;46:850–860. doi: 10.3109/10715762.2012.684878
38. Yoshida M, Ohkusa T, Nakashima T, Takanari H, Yano M, Takemura G, Honjo H, Kodama I, Mizukami Y, Matsuzaki M. Alterations in adhesion junction precede gap junction remodelling during the development of heart failure in cardiomyopathic hamsters. *Cardiovasc Res*. 2011;92:95–105. doi: 10.1093/cvr/cvr182
39. Lu X, Bi YW, Chen KB. Olmesartan restores the protective effect of remote ischemic perconditioning against myocardial ischemia/reperfusion injury in spontaneously hypertensive rats. *Clinics (Sao Paulo)*. 2015;70:500–507. doi: 10.6061/clinics/2015(07)07
40. Sun J, Nan G. The extracellular signal-regulated kinase 1/2 pathway in neurological diseases: a potential therapeutic target (review). *Int J Mol Med*. 2017;39:1338–1346. doi: 10.3892/ijmm.2017.2962
41. Bueno OF, Molkentin JD. Involvement of extracellular signal-regulated kinases 1/2 in cardiac hypertrophy and cell death. *Circ Res*. 2002;91:776–781. doi: 10.1161/01.res.0000038488.38975.1a
42. Muchir A, Shan J, Bonne G, Lehnart SE, Worman HJ. Inhibition of extracellular signal-regulated kinase signaling to prevent cardiomyopathy caused by mutation in the gene encoding A-type lamins. *Hum Mol Genet*. 2009;18:241–247. doi: 10.1093/hmg/ddn343
43. Muchir A, Wu W, Worman HJ. Reduced expression of A-type lamins and emerin activates extracellular signal-regulated kinase in cultured cells. *Biochim Biophys Acta*. 2009;1792:75–81. doi: 10.1016/j.bbadis.2008.10.012
44. Chiarini F, Evangelisti C, Cenni V, Fazio A, Paganelli F, Martelli AM, Lattanzi G. The cutting edge: the role of mTOR signaling in laminopathies. *Int J Mol Sci* . 2019;20:847. doi: 10.3390/ijms20040847

Downloaded from http://ahajournals.org by on February 4, 2026